Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / bending
Case	rca_5b9c29a9d4c8494089f9b0c89eb240d9
Date	2026-11-01 17:53:53
Status	AI proposal, pending expert review

WARNING: the AI model was not used for this analysis (fallback-fake (gemini failed: JSONDecodeError: Unterminated string starting at: line 56 column 9 (char 2567))). The text below is a generic offline template, not a result.

1. Problem

Defect	Severity	Confidence	Detail
crack	critical	95%	9.0 mm long

2. Root cause

Cause: Improper Bending Force / Over Bending

Reason: The detected deformation and dimensional inaccuracy strongly suggest excessive or improperly calibrated bending force. Retrieved RCA documents confirm this pattern for SPCC steel bending operations.

Effect: Deformation, Dimensional Inaccuracy

Confidence: 75%

Supporting evidence
- **seed-001**: Improper Bending Force or Over Bending causes deformation and dimensional inaccuracy. Precautions: optimize bending parameters, use proper tooling, calibrate machines regularly.
- **seed-004**: Springback in sheet metal bending causes dimensional inaccuracy. Overbend compensation and material property testing are required.

3. Solution

Corrective actions (fix this occurrence)

1. Re-set the press-brake bending force and angle to the material specification
2. Re-inspect the affected lot for deformation and dimensional deviation

Preventive actions (stop it recurring)

1. Add bending-force verification to the machine calibration schedule
2. Control bending parameters per material batch

Precautions

- Optimize bending parameters to material specification
- Use proper tooling and punch/die sets
- Regular machine calibration and force verification

How to verify the fix

Re-inspect the next 30 parts; no deformation and all dimensions within tolerance

4. Other causes considered

Rank	Cause	Confidence	Reason

2	Material Inconsistency / Hardness Variation	72%	Hardness variation in the incoming material can cause unpredictable springback and cracking. Material certificate verification is required.
3	Tool Wear / Die Damage	69%	Progressive die wear causes scratches, dents, and edge burrs. Preventive maintenance schedule should be implemented.

5. Traceability

AI model: fallback-fake (gemini failed: JSONDecodeError: Unterminated string starting at: line 56 column 9 (char 2567)). Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.